

Sara Edinger Murphy

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Education

California Institute of Technology Ph.D, Environmental Science and Engineering	Pasadena, CA August 2024
California Institute of Technology M.S., Environmental Science and Engineering	Pasadena, CA June 2021
Pomona College B.A., Chemistry (Minor in Linguistics) - <i>Cum Laude</i>	Claremont, CA May 2018

Professional Appointments

University of California, Los Angeles (Department of Chemistry) NSF Atmospheric and Geospace Sciences Postdoctoral Research Fellow	Los Angeles, CA September 2024 – present
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Teaching and Mentoring

Occidental College · Adjunct Professor · Physical Chemistry II (CHEM 305) · Foundations of General Chemistry Laboratory (CHEM 120L)	Fall 2023 - Spring 2024
UCLA Undergraduate Research Programs · Mentor · Mentored two undergraduate students during the 2024-2025 academic year · Designed and facilitated undergraduate research projects · Met one-on-one with students once a week to discuss progress and set goals · Supported the presentation of student work to the UCLA community	Fall 2024 - Present
Caltech Undergraduate Research Programs · Mentor · Designed research projects for undergraduates. · Trained students to operate laboratory instruments. · Trained students in skills essential for experimental atmospheric chemistry, including designing and executing experiments, data analysis, and literature searches. · Organized opportunities for students to present their work and contributions to the research group and to the wider Caltech community.	September 2020 - June 2022
California Institute of Technology · Teaching Assistant · Earth's Climate (ESE 1) · Quantum Chemistry (Ch 21a) · Earth's Biogeochemical Cycles (ESE 103) · Atmospheric Chemistry (ESE 171)	Spring 2020 - Spring 2023
Pomona College · Teaching and Laboratory Assistant · Physical Chemistry (Chem 158A PO) · General Chemistry (Chem 001 PO) · General Chemistry Laboratory (Chem 001L PO)	Fall 2015 - Spring 2017

Honors, Awards, and Grants

NSF Atmospheric and Geospace Sciences Postdoctoral Research Fellowship	2024–2026
Linde Laboratory Award for Community Building	2022
Received for excellence in service to the department and excellence in undergraduate mentorship.	
Rose Hills Foundation Graduate Fellowship	2018 - 2019
Richard Jahns Teaching Prize - <i>Nominated</i>	2021
Nominated by students for excellence in teaching during the pandemic	
Donald A. Strauss Fellow	2017 - 2018
Received a grant for \$10,000 to start and run a sustainable music program for underrepresented and low income elementary and middle school students in the city of Pomona at the dA Center for the Arts.	
Davis Projects for Peace Fellow	2016 - 2017
Received a grant for \$10,000 to run a summer music camp for low income elementary and middle school students in the city of Pomona.	

Publications

- **Sara E. Murphy**, John D. Crounse, Kristian H. Møller, Samir P. Rezgui, Nicholas J. Hafeman, James Park, Henrik G. Kjaergaard, Brian M. Stoltz, and Paul O. Wennberg. (2023). Accretion Product Formation in the Self-Reaction of Ethene-Derived Hydroxy Peroxy Radicals, *RSC Environmental Sciences - Atmospheres*.
- **Sara E. Murphy**, Reina S. Buenconsejo, Danielle C. Draper, John D. Crounse, Haroula D. Baliaka, Ryan X. Ward, Benjamin C. Schulze, Samir P. Rezgui, Katherine Ball, Tea Susskind, Gautham Kappaganthula, and Paul O. Wennberg. (2025). Multi-Functional Organic Nitrogen in the Los Angeles Air Basin, *ACS ES&T Air*.
- **Sara E. Murphy**, John D. Crounse, Andras Sun Poulsen, Juliette E. Lipson, Henrik G. Kjaergaard, and Paul O. Wennberg. (2025). Accretion Product Formation in the Self- and Cross-Reactions of Small β -Hydroxy Peroxy Radicals, *RSC Environmental Science - Atmospheres*.
- Paul Van Rooy, Afsara Tasnia, Barbara Barletta, Reina Buenconsejo, John D. Crounse, Christopher M. Kenseth, Simone Meinardi, **Sara Murphy**, Harrison Parker, Benjamin Schulze, John H. Seinfeld, Paul O. Wennberg, Donald R. Blake, and Kelley C. Barsanti. (2021). Observations of Volatile Organic Compounds in the Los Angeles Basin during COVID-19, *ACS Earth and Space Chemistry*.
- Lilian A. Dove, Clare E. Singer, **Sara E. Murphy**. (2024). Ten Steps to Make Qualifying Examinations in Geoscience Graduate Programs More Equitable, *AGU Advances*.

Presentations

- Caltech Environmental Science and Engineering Seminar, Pasadena, CA. *Formation of Accretion Products from Peroxy Radical Self-Reactions*. May 2020. Talk.
- Caltech Science and Society Seminar, Pasadena, CA. *Exploring alkene-derived peroxy radical chemistry in the lab and in the field*. October 2021. Talk.
- Atmospheric Chemical Mechanisms Conference, Davis, CA. *Formation of Accretion Products from Ethene-Derived Peroxy Radical Self-Reactions*. December 2022. Talk.
- Atmospheric Chemistry Gordon Research Conference, Newry, ME. *Identification and Quantification of Alkyl Nitrates in the Los Angeles Urban Atmosphere*. August 2023. Poster.
- Informal Gathering on Atmospheric Science and Photochemistry (IGASP), San Diego, CA. *Organic Nitrogen in the Urban LA Atmosphere*. October 2024. Poster.
- American Chemical Society (ACS) Meeting, San Diego, CA. *Structural Effects on Product Distribution in High RO₂ Systems*. March 2025. Poster.
- Atmospheric Chemistry Gordon Research Conference, Newry, ME. *Structural Effects on Product Distribution in High RO₂ Systems*. August 2025. Poster.

- American Chemical Society (ACS) Meeting, Atlanta, GA. *Observations of Organic Halogen Compounds During the Los Angeles Wildland-Urban Interface Fires*. March 2026. Talk.
- CIMS User Meeting, Billerica, MA. *Structural Effects on Product Distribution in High RO₂ Systems*. May 2026. Poster.

Invited Presentations and Guest Lectures

- Atmospheric Chemical Mechanisms Conference, Davis, CA. *The Formation of Accretion Products in the Self- and Cross-Reactions of Small Alkene-Derived Peroxy Radicals*. December 2024. Talk.
- Harvey Mudd College Hixon Center Chem and Climate Conversations, Claremont, CA. *Product Formation in the Self- and Cross-Reactions of Organic Peroxy Radicals*. April 2025. Talk.
- USC Earth Sciences Paleo/Environmental Seminar, Los Angeles, CA. *Bimolecular Peroxy Radical Chemistry in the Atmosphere*. October 2025. Talk
- PacifiChem, Honolulu, HI. *The Effect of Structure on Product Distributions in High RO₂ Systems*. December 2025. Talk.
- Harvey Mudd College, Claremont, CA. Guest Lecture in Chemistry of the Oceans and Atmosphere. April 2025.
- UCLA, Los Angeles, CA. Guest Lecture in Environmental Chemistry. June 2025.

Field Campaigns

Los Angeles Fire Study January 2025
 Deployed a PTR-ToF-MS at UCLA to examine the effects of the 2025 Los Angeles urban-wildland interface fires on local air quality.

Re-Evaluating the Chemistry of Air Pollutants in California (RECAP-CA) June - September 2021
 Deployed the Caltech GC-HRToF-CIMS, in conjunction with other instruments from Caltech and NOAA, to study changes in summer air quality in Los Angeles, specifically measuring hydroperoxides and nitrates.

Los Angeles Air Quality Campaign (LAAQC) June - September 2020
 Deployed the Caltech GC-HRToF-CIMS, in conjunction with other instruments from Caltech and UC Riverside, to observe changes in LA air quality during the pandemic.

Outreach and Community Engagement

Visiting Scientists January 2022 - September 2022
 Visited three Madison Elementary School 5th Grade classrooms for three hours each week to lead science lessons and hands-on activities for students.

Stars High School January 2021
 Designed and presented interactive, three-lesson course for high school students about the intersections of science and social justice.

Women in STEM Program August 2022
 Ran lab tours for high school women interested in pursuing STEM in college.

Draper Center for Community Partnerships September 2015 - May 2018
 Coordinated and led programs that promoted community outreach, including the Alternative Spring Break Program and the Next Level Program, a program designed to promote college access for local high school students in the city of Pomona.

Musician in Residence, dA Center for the Arts May 2016 - May 2018

Coordinated and taught music programs aimed at low income youth at the dA Center of the Arts in the city of Pomona.

Additional Skills

- Experience with the following instrumentation: Gas Chromatography, Chemical Ionization High Resolution Mass Spectrometry, Infrared Kinetic Spectroscopy, Excimer Lasers, UV/VIS Spectrometry, Liquid Chromatography Mass Spectrometry, FTIR
- Fixing and building instrumentation, including: soldering, designing and building circuits, editing and preparing instrument software, plumbing, leak checking
- Coding: Fluent in Matlab, Proficient in Python, Latex, and C++
- Grant Writing
- Organizing and leading meetings and programs
- Fluent in Spanish

References

- Paul O. Wennberg (wennberg@caltech.edu)
- Victoria P. Barber (vbarber@g.ucla.edu)
- Anne Yu (anneyu47@gmail.com)